V16

Rutherford Scattering

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1 Objective

The goal of the experiment is it to the Rutherford Atom modell. In addition is the goal to show proof the Ratherford scattering formular. Then will the thicness of the testet gold foil be meseaured thru the Bethe-Bloch equation.

2 Theoretical Background

For particle physics an imoportant concept is the cross section σ. The cross-section describes the propabylity, for a intaraction to happen. It can also be imagine as the effective target area for the pojectile. It is depended of the projectile, the energy of the projectile und the target, the unit of the cross-section is $1\,\mathrm{b} = 10^{-28}\,\mathrm{m}^2$. For this experiment the more important question is: where goes the scattered particle. This information is given in the differential cross section $\frac{\mathrm{d}\sigma}{\mathrm{d}\omega}$. For a nuclei which only has a Coulomb Potential it is given by the Rutherford formula, which will be handled in the next section.

2.1 Rutherford-scattering

The model the Rutherford experiment tried to proof is the so called "plum pudding" or Thompson model, wicht discribes the Atom as that the whole atom is evenly filled with positiv charges with the electrons inside of it. By this model only small scattering in small angles would be possible. But with the Rutherford scattering experiment it has been shown that this isn't the case. It showed that the porjectile whtich were α-particles are scattered in all direction, so big angles which shouldn't be possible. Rutherford's explanaiton is the so called Rutherford atom model which states, that only a very small positiv charged core is most of the mass and the electrons are around that.
For the scattering Rtutherford derived a formular which discribes the scattering. The assumptions of that are: that the the projectile and target are only point charges, the wohle force on the porjectile is only the coulomb force, the nuclei of the target is at rest and cannot move, it is calssicle and not relativistic and there is only one scattering and so no energy lost in the scattering. With this it rusults

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \left(\frac{1}{4\pi\epsilon_0}\right)^2 \left(\frac{zZe^2}{4E}\right)^2 \frac{1}{\sin^4(\theta/2)}\,.$$

There is z the nuclear charge number of the projectile und Z the one of the target. E is the kinetic energy of the projectile and θ the angle in wich the pojectile is beeing scattered [1].

2.2 Alpha decay and the ^{241}Am source

The alpha decay is defined as ${}^{\mathrm{A}}_{\mathrm{Z}}\mathrm{X} \rightarrow {}^{\mathrm{A}-4}_{\mathrm{Z}-2}\mathrm{Y} + {}^{4}_{2}\mathrm{He}^{++}$ where the ${}^{4}_{2}\mathrm{He}^{++}$ is the α particle itself. Americium-241 has been used as the α emitter. The half-life is $T_{1/2} = 432{,}6\,\mathrm{a}$ with an activity of $A = 330\,\mathrm{kBq}$ measured in October 1994, so the current activity is

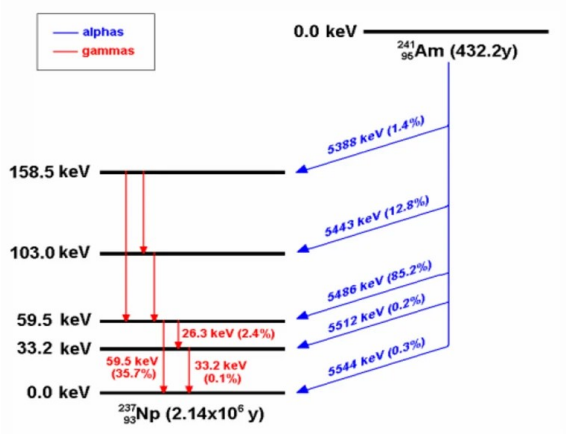


Figure 1: Decay scheme of ^{241}Am.

somewhat lower than that. The term diagram of ^{241}Am is given in Figure 1. The most important deacay is the one with an energy of 5,443 MeV. Also given are some γ decays with energys in the KeV range. So it is possible to diffentiat between the α and the γ decays.

2.3 Bethe-Bloch equation

Charged particles los energy if they go through material, this happens because of many small Coulomb collsions with atomic electrons, these ineratcation iosniese und excites the atoms. This energy loss is describes by the Bethe-Bloch equation

$$-\frac{\mathrm{d}E}{\mathrm{d}x} = \frac{4\pi e^4 z^2 N Z}{m_0 v^2 \left(4\pi\varepsilon_0\right)^2} \left[\ln\left(\frac{2m_0 v^2}{I}\right) - \ln\left(1-\beta^2\right) - \beta^2\right] \quad .$$

2.4 Surface barrier detector

3 Experimental Setup and Procedure

4 Analysis

The error calculation in this chapter is being done using the gaussian error propagation

$$\sigma_f = \sqrt{\left(\frac{\partial f}{\partial x_1}\sigma_{x_1}\right)^2 + \left(\frac{\partial f}{\partial x_2}\sigma_{x_2}\right)^2 + \dots}\,,$$

as well as the formulas

$$\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i\,,$$

for the mean and

$$\sigma_{\overline{x}} = \frac{\sigma_x}{\sqrt{n}} = \sqrt{\frac{1}{n(n-1)}\sum_{i=1}^{n}(x_i - \overline{x})^2}\,,$$

for the standard deviation. To simplify these calculations, the Python-package *uncertainties* [2] is being used. The absolute uncertainty of the counting rates, that were measured in this experiment are calculated with the formula

$$\Delta N = \sqrt{N} \implies \Delta A = \sqrt{\frac{A}{\Delta t}}\,,$$

because N is poisson-distributed.

4.1 Callibration of the Solid-Angle Measurement

To callibrate the zero point in the solid angle measuring apparatus, the measured counting rates from Table 1 are being shown in Figure 2.
The zero point $\Theta_0 = 0.5^\circ$ was selected, based on it being the highest measured count rate. Using *curve_fit* from *Scipy* [3] a gaussian fit was implemented and it shows that actually the most optimal Zero-Point would have been $\Theta_0^{\text{opt}} = 1.45^\circ$.

4.2 Examination of the Rutherford Scattering

The

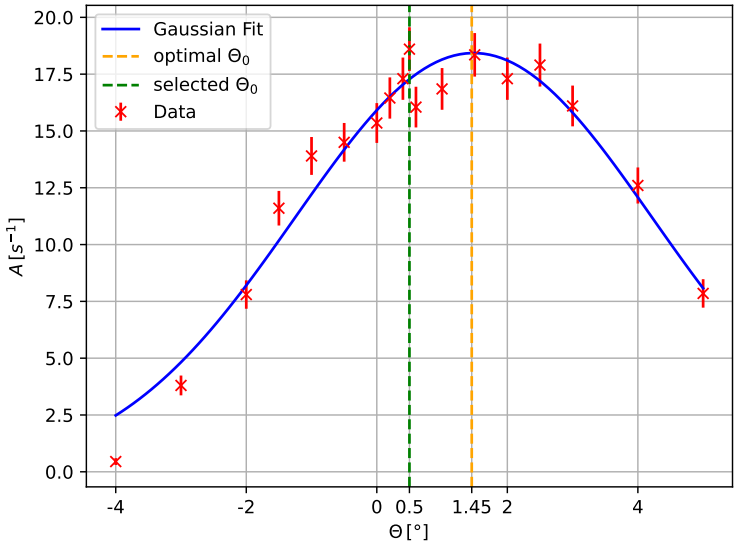


Figure 2: Measured count rates A plotted against the solid angle Θ. The selected Zero-Point Θ_0, as well as the optimal Θ_0 - based on a gaussian Fit - have been plotted as well.

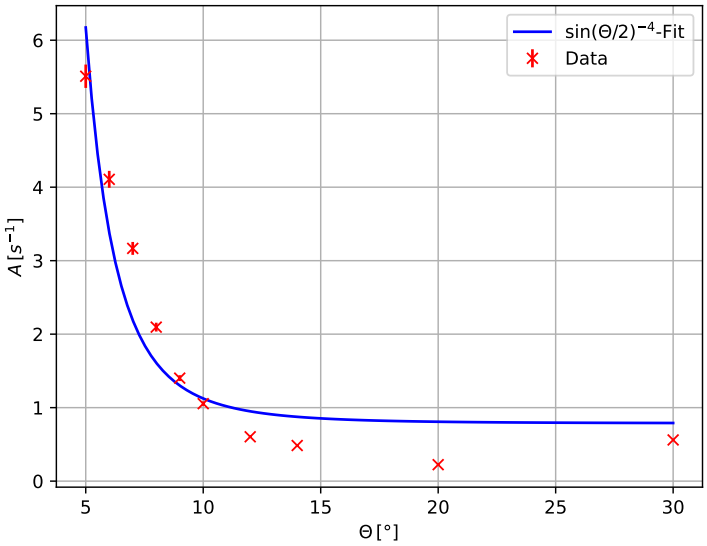


Figure 3: Measured count rates A plotted against the solid angle Θ.

5 Discussion

For the calculation of the relative experimental error, the formula

$$\Delta x = \frac{|x_{exp} - x_{theo}|}{x_{theo}},$$

is being used.

6 Attachment

Table 1: Measured Counts N in intervall time Δt for different angles Θ, as well as the calculated count rates A.

$\Theta\,[^\circ]$	N	$\Delta t\,[s]$	$A\,[s^{-1}]$
-4	9	20	0.45
-3	76	20	3.8
-2	156	20	7.8
-1.5	232	20	11.6
-1	278	20	13.9
-0.5	290	20	14.5
0	307	20	15.35
0.2	329	20	16.45
0.4	346	20	17.3
0.5	372	20	18.6
0.6	321	20	16.05
1	337	20	16.85
1.5	367	20	18.35
2	346	20	17.3
2.5	358	20	17.9
3	322	20	16.1
4	252	20	12.6
5	157	20	7.85

Table 2: Measured Counts N in intervall time Δt for different angles Θ, as well as the calculated count rates A.

$\Theta\,[^\circ]$	N	$\Delta t\,[s]$	$A\,[s^{-1}]$
20	760	3394.9	0.224
5	1202	218.16	5.51
6	1271	309.48	4.11
7	1305	412.06	3.17
8	1211	577.86	2.1
9	1237	882.56	1.4
10	1230	1166.8	1.05
12	1192	1975.1	0.604
14	1146	2364.6	0.485
30	37457	66844	0.56

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